

Utkarsh Pachapande

M.Tech Data Science Student | AI/ML | Generative AI | Data Analytics

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PROFESSIONAL SUMMARY

M.Tech Data Science student at PICT, Pune with hands-on experience building end-to-end Machine Learning, Deep Learning, NLP, and Retrieval-Augmented Generation (RAG) applications using Large Language Models (LLMs). Proficient in Python, SQL, TensorFlow, Scikit-learn, LangChain, Hugging Face, Power BI, and Streamlit. GATE Data Science & AI (DA) qualified (2024, 2025) and published researcher with two peer-reviewed papers in 2026, focused on building scalable, production-ready AI solutions.

EDUCATION

M.Tech in Data Science | Pune Institute of Computer Technology (PICT), Pune | SPPU 2025 – 2027

- CGPA: 9.55 / 10

B.E. in Artificial Intelligence & Data Science | Dr. D. Y. Patil Institute of Engineering, Management & Research, Pune | SPPU 2021 – 2025

- CGPA: 8.66 / 10

TECHNICAL SKILLS

Programming: Python, Java, C++, SQL

Machine Learning: Scikit-learn, TensorFlow, Keras, Pandas, NumPy, NLTK

Generative AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, LangGraph, Hugging Face, ChromaDB, FAISS, Ollama, Prompt Engineering

Data Analytics: Power BI, Tableau, Excel, Matplotlib, Seaborn

Tools & Web: Git, GitHub, Docker, Jupyter Notebook, VS Code, Streamlit, Flask, MySQL

EXPERIENCE

Data Science Intern | Ordnance Factory Varangaon (Munitions India Limited) Apr 2024

- Developed a real-time object detection system using YOLOv3 and OpenCV, capable of detecting 75+ object classes for defence-oriented surveillance use cases.
- Collaborated with a multidisciplinary engineering team to implement, test, and present computer vision solutions to technical stakeholders.

ML/AI Intern | ScaleFull Technologies LLP Dec 2023 – Jan 2024

- Built an end-to-end machine learning pipeline covering data preprocessing, feature engineering, model training, and evaluation.
- Deployed a Medical Insurance Cost Prediction system using Random Forest, ANN, and CNN architectures in TensorFlow, with a Streamlit front end for real-time predictions.

PROJECTS

RAG-Based Question Answering System | LangChain, FAISS, Ollama (Phi-3), Hugging Face, Streamlit Jan – Apr 2026

- Built a Retrieval-Augmented Generation (RAG) pipeline using LangChain, FAISS, MiniLM embeddings, LangGraph, and Ollama (Phi-3) for technical question answering over documentation.
- Evaluated retrieval and generation quality using the RAGAS framework, systematically reducing hallucinations and improving response relevance through iterative prompt and retrieval tuning.

Medical Insurance Cost Prediction | Random Forest, ANN, CNN, TensorFlow, Streamlit Dec 2023 – Jan 2024

- Developed and deployed a Streamlit-based insurance cost prediction system comparing Random Forest, Linear Regression, ANN, and CNN models for accuracy and generalization.

Object Detection using YOLOv3 | YOLOv3, OpenCV Apr 2024

- Built a real-time object detection system capable of identifying 75+ object classes using YOLOv3 and OpenCV.

Blinkit Sales Dashboard | Power BI, DAX Jan – Apr 2025

- Designed an interactive Power BI dashboard with KPI tracking and DAX measures to analyze and visualize sales performance trends.

CERTIFICATIONS

- [Introduction to Information Retrieval](#) — NPTEL (Elite + Gold)
- [Generative AI with LangChain & Hugging Face](#) — Udemy
- [Python for Data Science, AI & Development](#) — Coursera
- [The Data Science Course: Complete Data Science Boot Camp](#) — Udemy

ACHIEVEMENTS & PUBLICATIONS

- GATE Data Science & Artificial Intelligence (DA) Qualified — 2024, 2025
- Published "[Analysis of RAG-Based Question Answering Using Python Documentation](#)" — CEAJ, 2026
- Published "[Grocast: Interactive Sales Forecasting Dashboard](#)" — IRJAEH, 2025